WORKSHEET 9.8

Multiple-Choice Technique and the Last Two Weeks

Unit 9: Exam Strategy and Free-Response Types

Key ideas

- Eliminate first. A sign, a size or a unit often removes two options before any calculus.
- Do only the work the question needs. If the options differ in one feature, find that feature.
- Check an antiderivative by differentiating it, and a formula by trying an easy value.
- Two minutes without progress: choose, mark the question, move on. Never leave a blank.

Worked examples

Example 1 · Reading the leading terms

$\lim_{x\to\infty} \dfrac{3x^2+1}{5x^2-x}$ is

(A) 0 (B) $\dfrac{3}{5}$ (C) 1 (D) ∞

Solution. Equal degrees: the ratio of the leading coefficients.

(B).

Example 2 · Using symmetry

$\displaystyle\int_{-2}^{2} (x^3+1)\,dx$ is

(A) 0 (B) 2 (C) 4 (D) 8

Solution. The odd part x^3 integrates to 0 on a symmetric interval, leaving $\displaystyle\int_{-2}^{2} 1\,dx$.

(C).

Example 3 · Checking by differentiating

An antiderivative of $x\cos(x^2)$ is

(A) $\sin(x^2)$ (B) $\frac{1}{2}\sin(x^2)$ (C) $2\sin(x^2)$ (D) $-\frac{1}{2}\sin(x^2)$

Solution. Differentiate each option: $\frac{1}{2}\sin(x^2)$ gives $x\cos(x^2)$.

(B).

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Practice

No calculator on this page.

CORE

1. If $f(x) = \sin(3x)$, then $f'(0)$ is
   (A) 0 (B) 1 (C) 3 (D) -3

2. $\displaystyle\int_0^1 4x^3\,dx$ is
   (A) 1 (B) 4 (C) 12 (D) $\frac{1}{4}$

3. If $f'(x) < 0$ and $f''(x) > 0$, the graph of f is
   (A) increasing, concave up (B) decreasing, concave up (C) decreasing, concave down (D) increasing, concave down

4. $\displaystyle\lim_{x\to 0}\frac{\sin(5x)}{x}$ is
   (A) 0 (B) 1 (C) 5 (D) nonexistent

EXAM LEVEL

5. $\displaystyle\frac{d}{dx}\int_1^{x^2}\frac{1}{1+t^4}\,dt$ is
   (A) $\displaystyle\frac{1}{1+x^8}$ (B) $\displaystyle\frac{2x}{1+x^8}$ (C) $\displaystyle\frac{2x}{1+x^4}$ (D) $\displaystyle\frac{1}{1+x^4}$

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6. If $f'(x) = (x-1)^2(x-3)$, then f has
   (A) a relative minimum at $x = 1$ (B) a relative maximum at $x = 1$ (C) a relative minimum at $x = 3$ (D) a relative maximum at $x = 3$

7. The average value of $3x^2$ on $[0, 2]$ is
   (A) 4 (B) 8 (C) 12 (D) 2

8. The line tangent to $y = e^{2x}$ at $x = 0$ is used to approximate $e^{0.2}$. The approximation is
   (A) 1.02 (B) 1.1 (C) 1.2 (D) 1.4

Question 9 · Free-response practice

Answer each part in one or two sentences.

(a) Without evaluating it, explain why $\int_0^2 (x^2+1)\,dx$ cannot equal 1.

(b) A question offers xe^x+e^x and xe^x as possible derivatives of xe^x. Decide between them by considering the slope of the graph at $x = 0$, without the product rule.

(c) You have used 30 of the 38 minutes of the calculator multiple-choice part and answered 9 of 13 questions. Describe what you should do with the remaining time.

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Worked solutions

1. (C): $f'(x) = 3\cos(3x)$.
2. (A): $\left[x^4\right]_0^1 = 1$.
3. (B).
4. (C).
5. (B): substitute the upper limit and multiply by its derivative, $2x$.
6. (C): f' changes sign, from negative to positive, only at $x = 3$.
7. (A): $\frac{1}{2}\left[x^3\right]_0^2 = 4$.
8. (C): the tangent line is $y = 1 + 2x$, and $e^{0.2} = e^{2(0.1)}$, so use $x = 0.1$.
9. (a) On $[0, 2]$ the integrand is at least 1, so the integral is at least $1 \times 2 = 2$.
   (b) Near $x = 0$, $e^x \approx 1$, so $xe^x \approx x$ and the slope at $x = 0$ is 1. The first option gives $0 + 1 = 1$ and the second gives 0, so the first is correct.
   (c) About 2 minutes for each of the 4 remaining questions. Answer first those that need one calculator step, choose an option for every question, and leave none blank.

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